Introduction to Probability

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GENERAL ASSEMBLY

Data Science Process

1. Define problem.
2. Gather data.
3. Explore data.
4. Model with data.
5. Evaluate model.
6. Answer problem.

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Definitions

- **Experiment**: A procedure that can be repeated an infinite number of times and has a well-defined set of outcomes.

- **Event**: Any collection of outcomes of an experiment.

- **Sample Space**: The set of all possible outcomes of an experiment, usually denoted *S*.

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Examples

- **Experiment**: Flip a coin twice.
- **Sample Space**:
- **Event**:

- **Experiment**: Roll one die.
- **Sample Space**:
- **Event**:

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Definitions

- **Set**: An unordered collection of distinct objects.
  - { **Caroline, π, sweaterdresses** }

- **Element**: An object that is a member of a set.
  - **Caroline**
  - **π**
  - **sweaterdresses**

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Set Operations

- **Union**: A ∪ B = the set of elements in set A **or** in set B

- **Intersection**: A ∩ B = the set of elements in set A **and** set B

Example:

- A = {2, 4, 6, 8}
- B = {2, 3, 5, 7}

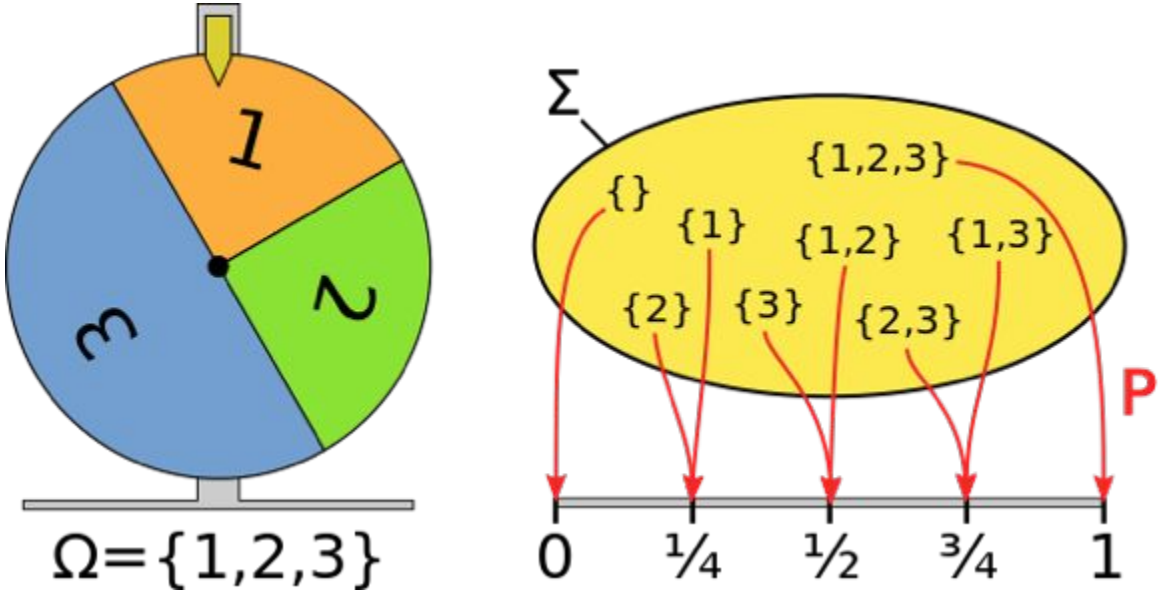
- A ∪ B =

- A ∩ B =

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Probability

We are often interested in the probability of some event(s) occurring.



We write P(A) to mean the probability that event A occurs.

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Probability - Practice

- A: “a U.S. birth results in twin females”
- B: “a U.S. birth results in identical twins”
- C: “a U.S. birth results in twins”

- In words, what does $P(A \cap C)$ mean?

- In words, what does $P(A \cap B \cap C)$ mean?

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Example adapted from *Statistical Inference*, 2nd Ed., Casella & Berger

Probability Rules

When trying to find the probability of a complex event, it's not straightforward.

- There are 12 red and 12 black balls. If you draw one ball, then a second ball without replacing the first, what is the probability they are the same color?
- Suppose you roll three dice. What is the probability that the three dice are rolled in increasing order?
- You call 3 friends of yours in Seattle and ask each independently if it's raining. Each of your friends tells you the truth ⅔ of the time. All 3 friends tell you it is raining. Based on historical evidence, it rains ¼ of the time in Seattle. What is the probability that it's actually raining in Seattle right now?

There are three probability rules that come in handy.

Probability Rule 1: P(A ∪ B)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- Venn diagrams can help *illustrate* this, but Venn diagrams are not formal proofs!

Probability Rule 2: P(A | B)

$$P(A \mid B) = P(A \cap B) / P(B)$$

- A | B means “A given B” or “A conditional on the fact that B happens.”

Probability Rule 3: P(A ∩ B)

$$P(A \cap B) = P(A \mid B) * P(B)$$

- We just took the last rule and multiplied both sides by P(B).

- We can rearrange these as well: $P(B \cap A) = P(B \mid A) * P(A)$

- This isn't limited to two events: $P(A \cap B \cap C) = P(A \mid B \cap C) * P(B \mid C) * P(C)$

Probability Practice

- There are 12 red and 12 black balls. If you draw one ball, then a second ball without replacing the first, what is the probability that they are the same color?

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When by hand is tough...

- Oftentimes, it's challenging to evaluate probabilities by hand.

- But it's important to understand the ideas behind probability.
  - For example: are two events independent of one another?
  - Events A and B are **independent** if **P(A | B) = P(A)**.

- We often think of probability as how frequently an event occurs.
  - We can use computer simulations to give us a good approximation of the true probability of some event.
  - We can often use this to "check our work" or to tackle more challenging probability problems!

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